* Revision:

* feature Enginneering:

→

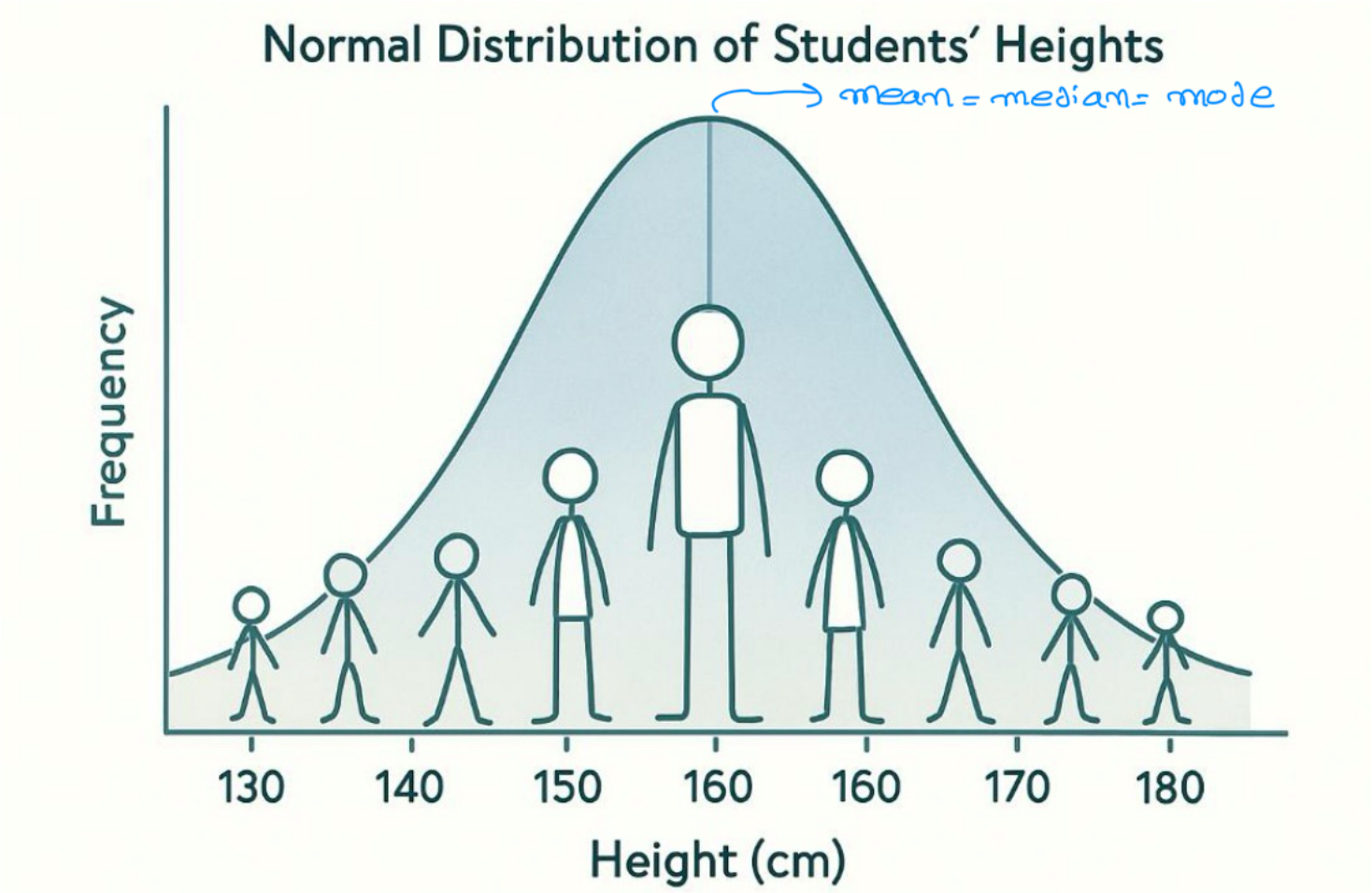
→ Pending: transformers & feature scaling -

* Stistics:

- L12/L13 : PMF & PDF

① Normal distribution:

- Bell shaped curve & symmetric around mean.


Normal Distribution of Students' Heights
mean = median = mode
Frequency
130
140
150
160
160
170
180
Height (cm)

desmos.com

0 5 10 15 20 25 30

$$\mu = \frac{0+5+10+15+20+25+30}{7} =$$

σ: spread 0 5 10 15 20 25 30

$$\sigma^2 = \frac{\Sigma (x-\mu)^2}{n} = \frac{(0-15)^2+(5-15)^2+(10-15)^2+(15-15)^2+(20-15)^2+(25-15)^2+(30-15)^2}{7}$$

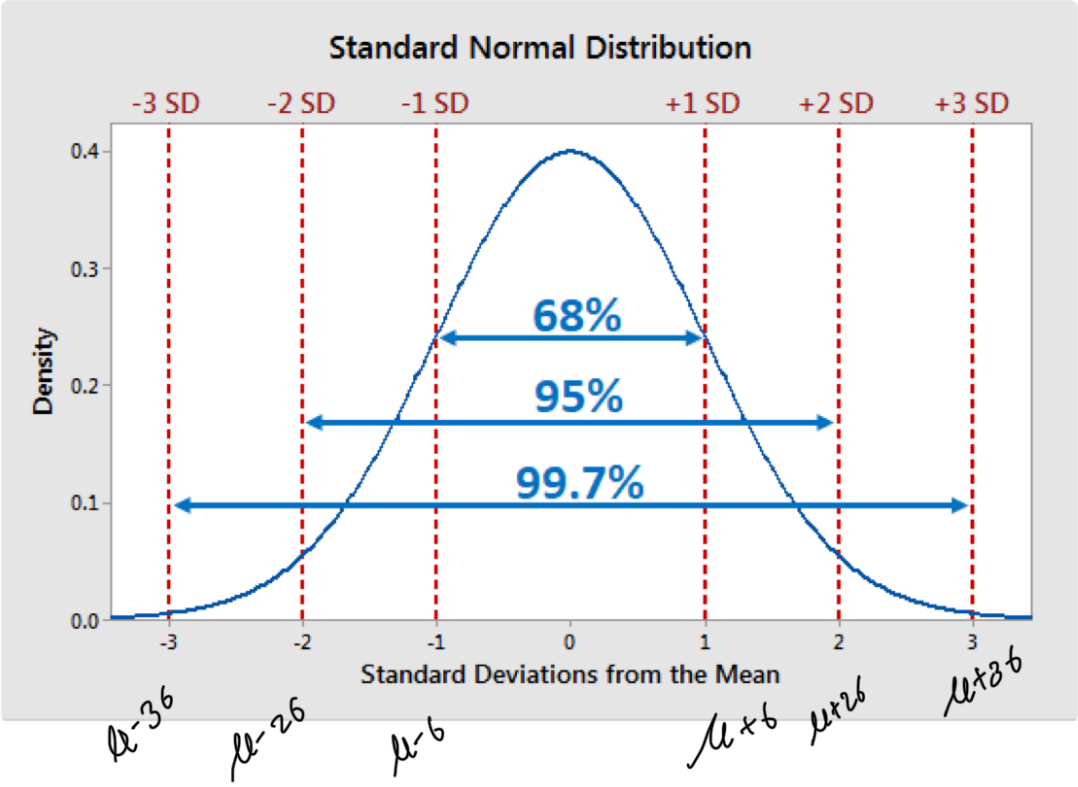
$$= \frac{(-15)^2+(-10)^2+(-5)^2+0+(5)^2+(10)^2+(15)^2}{7}$$

$$= \frac{225+100+25+25+100+225}{7}$$

$$\sigma^2 = \frac{700}{7} = 100$$

$$\sigma = \sqrt{\sigma^2} = \sqrt{100} = 10$$

* Empiricel Rule:



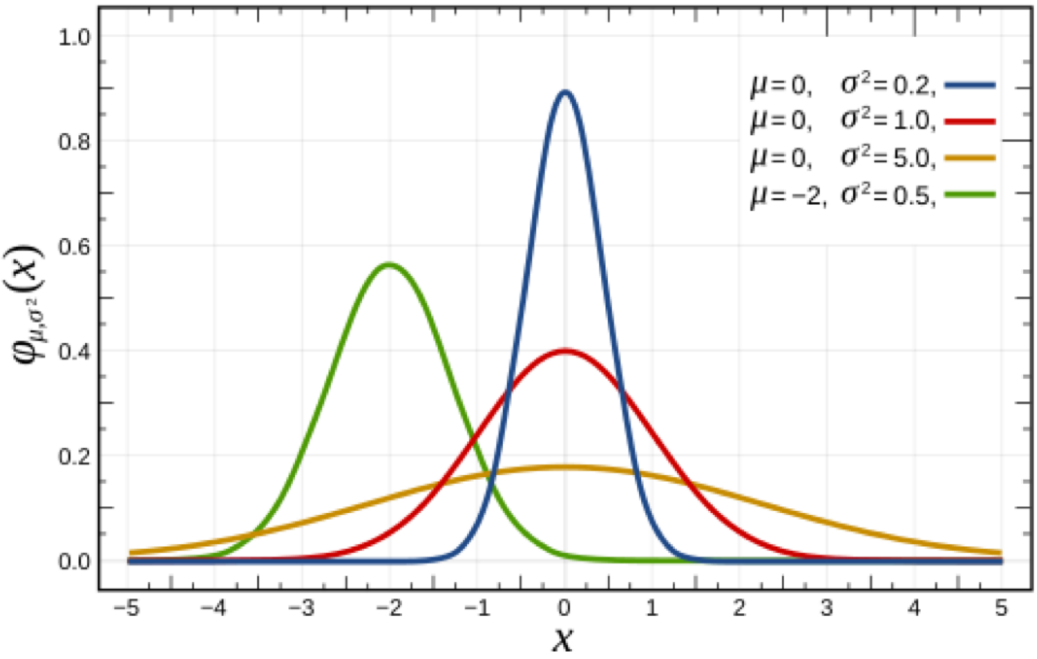
→ If we know μ & σ of any dataset & we also know it is normally distributed then we can plot the distribution.

→ $$y = \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2\left(\frac{x-\mu}{\sigma}\right)^2}$$

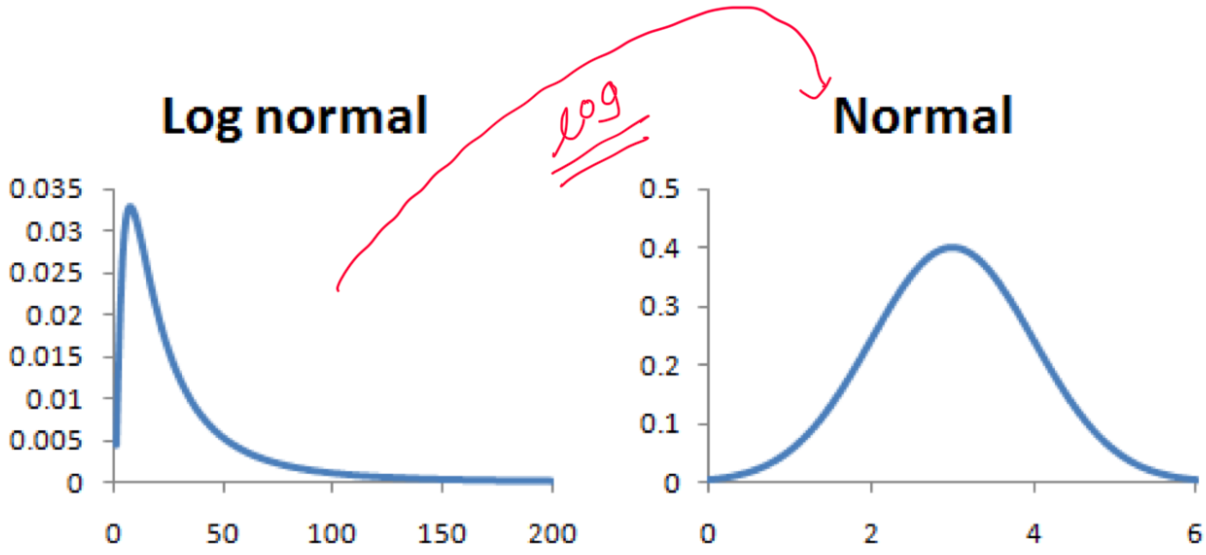
⇒ Properties of Normal distribution:

- Symmetric around it's mean.
- mean = median = mode
- Empirical Rule
- Area under the curve = (?)
- parameters: (μ, σ)

$$X \sim N(\mu, \sigma)$$



* Log Normal distribution:



* Prob. dis. whose logarithm is normally distributed.

$X \sim \log\text{normal}(\mu, \sigma)$

$\log(X) \sim N(\mu', \sigma')$

Eg: Income of the people. & outlier

Usecase: convert Right skewed data → normal dis.

* z score:

- How many std a data point lies away from the mean.

Eg: 0 5 10 (15) 20 25 (30)

1 +0.5

$\mu = 15$

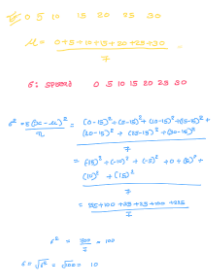
$\sigma = 10$

$x = 30$ } 1.5 std

$x = 35$? 2 std

$35 - 15 =$

$\frac{(20)}{6} \Rightarrow \frac{20}{10} =$ 2 std



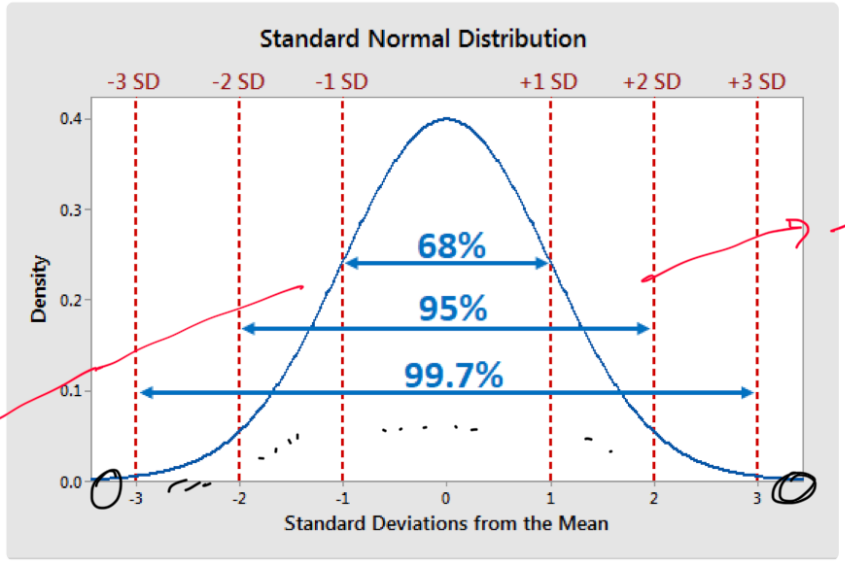
$x = 40$: $x - \mu \Rightarrow 40 - 15 = \frac{25}{6} = \frac{25}{10} = 2.5$ std

→ $x = -5$ $\frac{-5-15}{10} = \frac{-20}{10} = -2$ std

→ $z\ score = \frac{x - \mu}{\sigma}$ z score of μ = 0

* what can we know from z score ?

-



→ If $|z| > 3$ → very less proba

$z = 0.5$ → high proba.

→ we can compare diff. data points.

* Z-score method to detect outliers :

if $|z| > 3$ → outliers

* Standard Normal distribution:

– Normal dist. with $\mu = 0$ & $\sigma = 1$.

Suppose two colleges conduct different exams. We want to compare the performance of two students.

Details	College A	College B
Class average (mean)	70	50
Standard deviation	10	5
Student's marks	85	60

Question: Which student performed better compared with their own classmates?

→ A: $z = \frac{x - \mu}{\sigma} = \frac{85 - 70}{10} = \frac{15}{10} = 1.5$

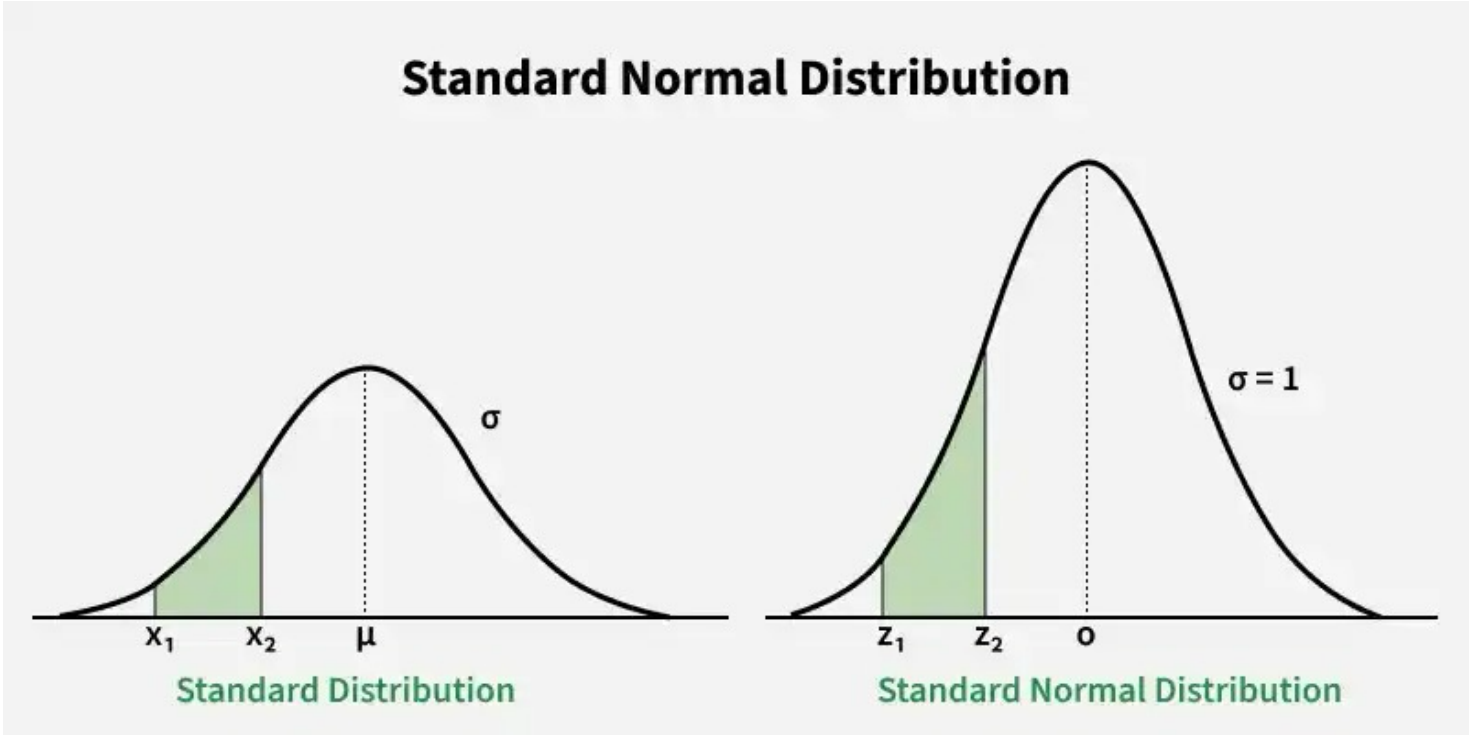
B: $z = \frac{x - \mu}{\sigma} = \frac{60 - 50}{5} = \frac{10}{5} = 2$

"Raw marks alone may not tell us how well a student performed relative to their classmates. By converting marks into z-scores, we measure how far each student is from their own class average in standard deviation units."

→ convert og value → z-score } For normally distributed data

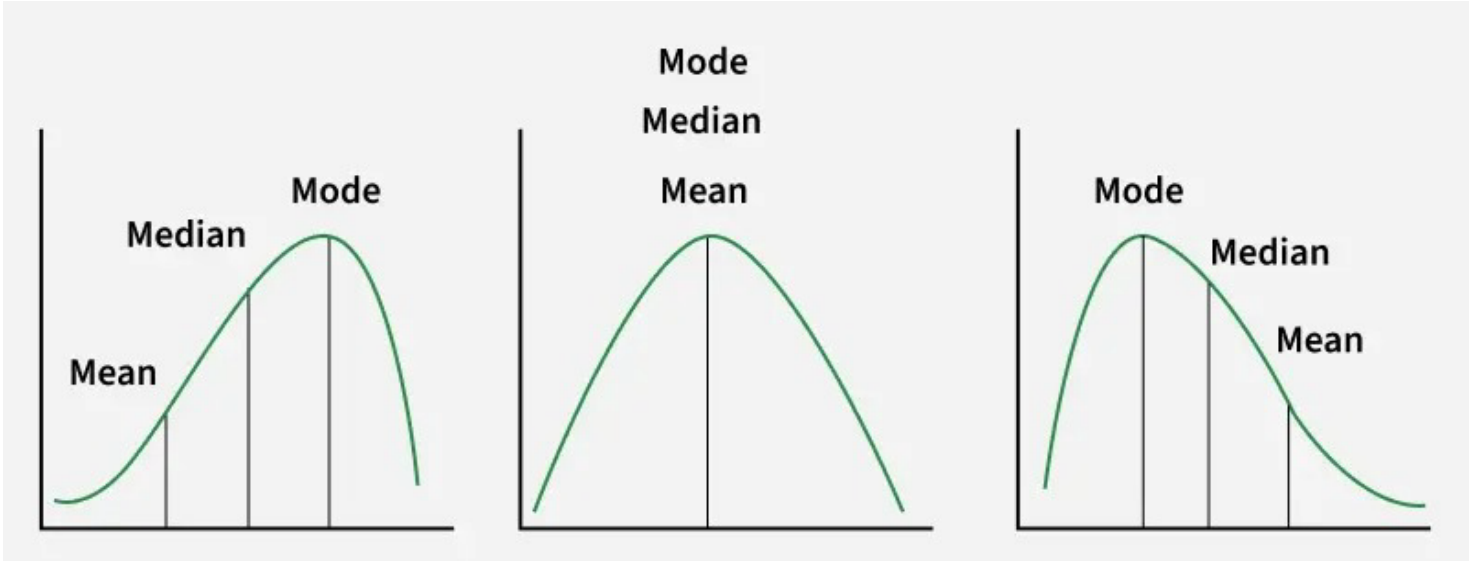
it gives $\mu=0$, $\sigma=1$ } std normal dis.

$X \sim N(0,1)$



→ we can convert diff. dist. to the same standard scale for comparision.

* Skewness:


Mean
Median
Mode
Mode
Median
Mean
Mode
Median
Mean